

Practical Lightweight-Cipher Construction under a Time- and Budget-Bounded Field Adversary

Equal-Margin Software Measurements, the AC/DC Objective, and a DONATION Case Study

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Abstract

Lightweight block-cipher design is usually discussed in terms of area, latency, throughput, or energy on a selected implementation platform. For a short-lived field communication channel, however, the relevant design problem is asymmetric: the legitimate endpoint is constrained to a low-tier general-purpose processor, whereas a well-organized adversary can aggregate substantially more capable software and hardware resources at an interception point. This paper develops a practical comparison methodology for that setting. The operational scenario is a four-hour, 12 Mbit/s video stream protected by one mission key, with an adversarial equipment budget of 1 million USD per interception point and an acceptable probability of a useful compromise of at most 10^{-6} . Cipher round functions are first normalized by a common preliminary criterion: r_{64} is the smallest round count for which every single differential trail has probability at most 2^{-64} , and each implementation is measured at $X = r_{64} + 1$. All practical measurements are performed on the same high-performance general-purpose CPU and reported in nominal invariant-TSC ticks per byte. The platform intentionally represents neither the constrained legitimate endpoint nor the attacker’s specialized setup; it is used only for controlled relative comparison. The results show that round-function structure dominates round count. At the common yardstick, PRESENT with the `lin444` (2, 9, 7) layer is the fastest measured construction at $X = 7$ and 0.502 cyc/B, approximately 9% below reduced PRESENT-80 at $X = 16$. An unstructured 8-bit S-box makes the original DONATION construction table-bound and comparatively expensive. Replacing DONATION’s transpose by an XOR-and-rotate layer substantially improves its single-trail differential bound at negligible table-driven cost, and replacing its S-box by the structured AES S-box enables a compact bitsliced implementation. For 128-bit blocks, reduced AES and an AES-S-box construction with the XOR-and-rotate layer are approximately tied within measurement uncertainty. The broader design objective is to maximize the attacker-cost to defender-cost ratio (AC/DC): ideally, a construction change should increase the resources required for a successful attack without increasing the cost of protecting the mission stream. The present data determine the defender-side software ordering and provide source-backed adversarial anchors, but do not yet support a numerical AC/DC ranking for the exact reduced variants. The paper therefore identifies the construction approach that currently looks best under the measured software and differential-trail model, quantifies a

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source-backed GPU reference adversary, and separates measured attack rates from equipment and hardware-scaling assumptions.

Keywords: lightweight cryptography; differential trail; block cipher; implementation cost; time-bounded adversary; PRESENT; AES; DONATION; bitslicing.

1 Introduction

“Lightweight” is not a scalar property. A bit permutation can be almost free as wiring in hardware and as register renaming in a bitsliced implementation, yet expensive in a direct scalar implementation. A byte-sized substitution table can be convenient on a conventional processor, but incompatible with strict constant-time execution and potentially expensive as a Boolean circuit. Conversely, a highly structured S-box may admit specialized instructions or a compact published circuit that benefits an attacker at least as much as the legitimate user. The optimization target therefore has to be stated together with the operational model and the implementation family.

The motivating application is real-time unmanned-platform video. The current DONATION specification is an unpublished manuscript that targets online UAV video encryption; an earlier workshop publication and a preceding temporary-security proposal state the same general motivation [1, 2, 3]. The current manuscript, rather than the earlier printed S-box, is the normative source for the implementation studied here. The present study does not assume that its performance and security claims are either correct or incorrect. Instead, it asks a broader engineering question:

Which block-cipher construction approach minimizes defender-side cost while maximizing the resources required by a scalable, time- and budget-bounded attacker?

The work makes four contributions.

1. It frames the central design objective as the attacker-cost/defender-cost ratio (AC/DC), rather than defender throughput alone, and states precisely why the current data are sufficient only for a partial evaluation of that objective.
2. It states an operational adversary model for a short-lived field channel and defines a reproducible preliminary comparison rule based on the minimum weight of a single differential trail.
3. It compares several representative round-function architectures after applying the same set of applicable software optimizations and reducing each to the same differential-trail margin.
4. It uses DONATION and controlled modifications of DONATION as an application-oriented case study, including negative findings where the evidence requires them.

The paper is deliberately preliminary. It does not claim complete cryptanalysis of reduced-round designs, and it does not claim to have measured the actual transmitter or the actual attack cluster. Remaining publication metadata and optional validation work are marked explicitly rather than replaced by optimistic assumptions.

2 Operational scenario and adversary model

2.1 Mission profile

The current scenario parameters are summarized in table 1. One key is provisioned for the entire mission. No online rekeying is assumed. The protected data are a parallelizable video stream; occasional frame loss is acceptable. The reference mode is therefore counter-mode-like encryption with rigorously unique counter inputs. Authentication is outside the present model: if the adversary can actively affect the radio channel, denial or jamming is considered a more direct

operational action than subtle ciphertext manipulation. This is a scope decision, not a general recommendation to omit integrity protection.

Table 1: Operational scenario used in the draft.

Parameter	Assumption
Useful confidentiality interval	$T = 4 \text{ h} = 14,400 \text{ s}$
Video rate	$R = 12 \text{ Mbit/s}$
Data per mission	$D = RT = 172.8 \text{ Gbit} = 21.6 \text{ GB}$
Keying	One mission key; no in-mission rekeying
Connections	A small number of point-to-point channels
Mode	Parallelizable counter-mode-like encryption with unique counters
Adversary budget	$B = 1 \text{ million USD}$ per interception/attack point
Target compromise probability	$\varepsilon = 10^{-6}$ per operation
Excluded	Quantum computation, physical key extraction, side channels, active-channel cryptanalysis, and related-key analysis

The amount of processed data is

$$D = RT = 12 \times 10^6 \times 14400 = 1.728 \times 10^{11} \text{ bits.} \quad (1)$$

For block size n , the number of block-cipher calls is

$$q_n = \frac{D}{n}. \quad (2)$$

Hence

$$q_{64} = 2.7 \times 10^9 \approx 2^{31.33}, \quad (3)$$

$$q_{128} = 1.35 \times 10^9 \approx 2^{30.33}. \quad (4)$$

2.2 Time- and budget-bounded field adversary

The adversary is an organization acting at a field interception point. It can deploy a network of sniffers, aggregate traffic from several nearby targets, prioritize high-value channels, and scale any useful software or hardware implementation. It cannot economically assign an unlimited supercomputer to every channel; the budget is local to one attack point and must be divided among the simultaneously observed targets. This framing follows the concrete-security tradition, where adversarial advantage is expressed as a function of time, data, and other resources [4, 5, 6].

Let

$$\mathcal{A}_{\text{op}} = (T, B, \mathcal{P}, D, M) \quad (5)$$

characterize an operational adversary with attack time T , equipment budget B , available platform classes $\mathcal{P}$, intercepted data D , and up to M observable targets. The success criterion is

$$P_{\text{op}}(C) = \Pr[\text{at least one operationally useful compromise of } C \text{ before } T] \leq \varepsilon, \quad (6)$$

where $\varepsilon = 10^{-6}$.

For ordinary exhaustive search with a k -bit effective key and a total budget of Q_C complete candidate checks at one attack point,

$$P_{\text{key}}(C) \approx \min\left(1, \frac{Q_C}{2^k}\right). \quad (7)$$

The number of targets does not independently multiply Q_C when Q_C already denotes the total work available at the point. How the attacker schedules that work is part of the strategy.

The resource model deliberately separates four evidence classes: vendor specifications, published complete-key-search measurements, arithmetic scaling of independent devices, and unverified engineering projections. Only the first three are used for numerical conclusions.

2.3 Attacker-cost/defender-cost objective

The main design objective is not merely to minimize encryption time. Let $C_D(C)$ denote the defender’s cost of protecting the fixed mission workload with construction C , and let $C_A(C; T, \varepsilon)$ denote the minimum adversarial cost required to reach success probability ε within the useful attack interval T . When both costs are expressed on a common accounting basis, the direct ratio is

$$\text{AC/DC}(C) = \frac{C_A(C; T, \varepsilon)}{C_D(C)}. \quad (8)$$

For cross-platform comparisons, a dimensionless form relative to a reference construction C_0 is more practical:

$$\text{AC/DC}_{C_0}(C) = \frac{C_A(C; T, \varepsilon)/C_A(C_0; T, \varepsilon)}{C_D(C)/C_D(C_0)}. \quad (9)$$

A value above one means that, relative to C_0 , the construction shifts cost against the attacker; a value below one shifts cost in the attacker’s favor. The ideal design change raises attacker cost while leaving defender cost effectively unchanged—an AC/DC gain obtained “for free” on the legitimate path.

The present experiment does not produce a numerical value of equation (9). It measures a controlled software proxy for C_D and imports complete-key-search anchors for full AES and PRESENT, but it has no GPU, FPGA, or ASIC implementations of the exact reduced variants and no common accounting model for structural attacks. AC/DC is therefore the main question raised by the paper and the logical target of the next experimental stage, not a result claimed from the current data.

2.3.1 Equipment envelope

Table 2 shows configurations that fit approximately within the per-point capital limit. They are not asserted to be optimal procurements. The CPU count follows from Intel’s published price, core count, and two-socket support for the Xeon 6980P [7]. The GPU configuration is a scenario assumption of \$5000 per installed accelerator, including its share of host, power delivery, cooling, storage, and networking. This is deliberately much higher than the RTX 4090 launch price of \$1599 [8], and leaves ten percent of the budget unallocated. The FPGA count follows directly from AMD’s published Alveo V80 MSRP; AMD specifies 2.6 million LUTs and a maximum board power of 190 W [9]. No current per-cipher V80 implementation is available, so the device count must not be converted into a key-search rate by LUT-proportional scaling.

2.3.2 Published GPU reference case

The strongest directly usable evidence is complete candidate-key throughput measured on one RTX 4090. Tezcan and Leander report $2^{34.64}$ AES-128 keys/s and $2^{32.90}$ PRESENT-80 keys/s, including generation of the candidate-dependent round keys [10]. The AES value extends an earlier measured result of $2^{32.433}$ keys/s on an RTX 2070 Super [11]; the PRESENT lineage includes a measured 1.89×10^9 keys/s on an RTX 3070 [12]. Because exhaustive-search partitions are

Table 2: Equipment envelope for one \$1M attack point. Counts are budget-feasible scenarios, not measured cryptanalytic clusters.

Class	Assumed count	Direct component cost	Status and limitation
CPU	32 Xeon 6980P processors	\$446,560	4096 physical cores in 16 possible two-socket nodes; complete-node cost and per-cipher candidate rate are not measured
GPU	180 RTX 4090-class accelerators	\$900,000	\$5000 installed-cost assumption; published per-card key-search rates are available
FPGA	90 Alveo V80 cards	\$854,550	Vendor price and resources are known; no present-work cipher cores were synthesized
ASIC	—	—	Considered only when a completed design already exists; NRE, node, yield, packaging, and volume prevent a defensible universal count

independent, scaling to $N_G = 180$ accelerators for $T = 14400$ seconds gives

$$Q_C^{\text{GPU}} = N_G T R_C. \quad (10)$$

The resulting four-hour capacities are shown in table 3.

Table 3: Source-backed GPU exhaustive-search reference for one attack point. These rates apply to the full published ciphers, not automatically to the reduced experimental variants.

Cipher	One RTX 4090	180 GPUs, 4 h	P_{key} for stated key	Evidence
AES-128, 10 rounds	$2^{34.64}$ keys/s	$2^{55.95}$ keys	$2^{-72.05}$	Published measured kernel
PRESENT-80, 31 rounds	$2^{32.90}$ keys/s	$2^{54.21}$ keys	$2^{-25.79}$	Published measured kernel

For an acceptable exhaustive-search success probability ε , the effective key entropy must satisfy

$$k \geq \log_2 Q_C + \log_2 \left(\frac{1}{\varepsilon} \right). \quad (11)$$

With $\varepsilon = 10^{-6}$, the measured GPU reference requires approximately 76 effective bits for AES and 75 bits for PRESENT. A deliberately conservative generic envelope of $Q_C = 2^{60}$ requires 80 bits, while even $Q_C = 2^{63}$ requires 83 bits. Thus a 96-bit effective key has substantial reserve against exhaustive search in this model. This conclusion does not transfer to structural attacks with lower complexity.

The equal-margin experimental variants use fewer rounds than the published ciphers. A purely inverse-round-count extrapolation would give upper envelopes of at most $2^{56.95}$ checks for AES-r5 and $2^{55.16}$ for PRESENT-r16. These are not measurements: candidate-key setup, memory

behavior, and fixed overhead do not scale linearly with round count. They are included only to show that the key-entropy conclusion is insensitive to a one-bit change in estimated work.

2.3.3 FPGA and ASIC interpretation

A historical hardware anchor exists for PRESENT. Pospisil and Novotny implemented 240 pipelined PRESENT breakers on the 120-Spartan-3 COPACOBANA platform and measured 24 billion complete keys/s [13]. This corresponds to $2^{48.30}$ candidates in four hours for one system. The original COPACOBANA platform was reported as costing less than \$10,000 [14]; a purely historical cost replication of 100 such systems would reach about $2^{54.94}$ candidates. This is evidence that PRESENT maps efficiently to specialized hardware, but it is not a 2026 V80 prediction. Process technology, routing, host cost, and implementation strategy invalidate direct LUT or frequency scaling.

No equally grounded FPGA result exists for the supplied DONATION S-box, `lin444` variants, or reduced AES configurations. The earlier project ranges of roughly 2^{54} – 2^{60} FPGA evaluations and 2^{57} – 2^{63} pre-existing-ASIC evaluations are retained only as sensitivity envelopes in the discussion, not as measured results. ASIC development cost is omitted from the scenario as agreed, but this does not make manufacturing throughput technology-independent.

For DONATION, the supplied key is a raw 576 bit string split into nine independent 64 bit round keys. There is no key schedule. Exhaustive key search is therefore outside the practical model of this case study; the binding questions are structural cryptanalysis, data complexity, and round-function cost.

2.4 Implications of a 64-bit block

A 64-bit block is not automatically broken after 2^{32} encrypted blocks. The exact consequence depends on the mode. In CBC-like modes, birthday collisions can be converted into practical plaintext recovery when part of the plaintext is predictable or attacker-controlled, as demonstrated by Sweet32 [15]. In CTR mode with unique counter inputs T_i ,

$$C_i = P_i \oplus E_K(T_i), \quad (12)$$

and the permutation property implies that distinct T_i produce distinct keystream blocks. The CBC collision equation does not apply. Counter reuse, however, is catastrophic:

$$C_i \oplus C'_i = P_i \oplus P'_i. \quad (13)$$

NIST therefore requires counter blocks to be unique under a given key [16].

2.4.1 Empirical avalanche sanity check

The supplementary `avalanche.csv` artifact records a cellwise bit-independence experiment for the ciphers listed in that file. For each input bit i and output bit j , the experiment toggles input bit i and estimates the probability that output bit j changes. The supplied run uses 10^6 trials per cell, seed `0x00001ad46a72c572`, and an acceptance tolerance of $\varepsilon_{av} = 0.01$ around the ideal rate $1/2$; the supplied result marks the tested cells as passing.

This experiment has a deliberately narrow interpretation. For CTR with distinct counter blocks, non-repetition of keystream blocks follows deterministically from the permutation property:

$$T_i \neq T_j \implies E_K(T_i) \neq E_K(T_j). \quad (14)$$

Avalanche behavior is not needed to establish that fact. The experiment instead checks for conspicuous local output-bit bias when two cipher inputs differ in one bit. Adjacent integer counters are numerically close but do not always differ in exactly one bit, because carry propagation changes their Hamming distance. The test does not prove pseudorandomness, bound differential probabilities, exclude structural attacks, or prevent two ciphertext blocks from coinciding after different plaintext blocks are XORed with distinct keystream blocks.

The generic PRP/PRF switching term after q queries is bounded by

$$\Delta_{\text{switch}} \leq \frac{q(q-1)}{2^{n+1}} \quad (15)$$

for an n -bit block [17, 18]. At the current mission volume,

$$\Delta_{64} \lesssim 0.198, \quad (16)$$

$$\Delta_{128} \lesssim 2.68 \times 10^{-21} \approx 2^{-68.34}. \quad (17)$$

The first value is a generic distinguishing bound, not a demonstrated video-recovery probability. It nevertheless shows that 64-bit block ciphers cannot support a strong generic indistinguishability statement for the full mission. Therefore the paper reports two distinct conclusions for every 64-bit design: primitive-level differential-trail behavior and mode-level applicability. A 64-bit design is not rejected solely by block size, but its use is conditional on unique-counter management and acceptance of the generic birthday-scale limitation. The withdrawal of TDEA as an approved NIST block cipher is consistent with the broader migration away from new 64-bit-block deployments [19].

3 Comparison methodology

3.1 Common differential-trail yardstick

For cipher C , let $W_C(r)$ denote the minimum negative binary logarithm of the probability of any nonzero r -round differential trail:

$$W_C(r) = \min_{\tau \in \mathcal{T}_r(C)} -\log_2 \Pr[\tau]. \quad (18)$$

Define

$$r_{64}(C) = \min\{r : W_C(r) \geq 64\}, \quad (19)$$

and benchmark the cipher at

$$X(C) = r_{64}(C) + 1. \quad (20)$$

For a 64-bit block, 2^{64} is the full-codebook scale, which makes $W = 64$ a natural hard lower bound for the present screening experiment. For 128-bit ciphers it is only a common cross-design yardstick, not the codebook complexity. The paper may reconsider the threshold if the final operational-adversary estimate contradicts it.

The SAT decision procedure asks whether an r -round trail with weight at most 63 exists. SAT returns a witness, which is replayed against the cipher model. UNSAT proves $W_C(r) \geq 64$. This direct decision procedure is tighter and cheaper than extrapolating a deeper-round bound from disjoint windows.

The criterion covers only a single differential trail under independent round keys. It does not prove resistance to differentials, differential clustering, linear cryptanalysis, integrals, algebraic attacks, related-key attacks, or complete key recovery. In particular, X is a comparison instrument, not a proposed production round count.

3.2 Fair optimization rule

A comparison at designed round counts answers a different question: it mixes round-function cost with the designers’ chosen security margin. We therefore reduce all compared constructions to equation (20) and apply every predefined optimization family that is structurally applicable.

The tested software families are:

1. scalar reference implementation;
2. fused substitution/linear-layer tables;
3. interleaved two-, four-, eight-, and sixteen-block table kernels where available;
4. scalar bitslice;
5. scalar bitslice with transposition moved outside the timed kernel;
6. AVX2 bitslice;
7. AVX2 bitslice with bitsliced state at the boundary.

Hardware finite-field acceleration is disabled: AES-NI, GFNI, and VAES are not used. The fastest verified kernel is selected separately for each cipher. Forcing one implementation family on every construction would itself be an unfair restriction.

This rule is essential. In the PRESENT/lin444 experiment, specializing the modified linear layer before equivalently optimizing PRESENT’s original permutation initially produced a misleading break-even result. After both sides received comparable optimization effort, the equal-margin comparison moved by roughly 9–16% against the modified layer. Thus the object being compared is always a pair of optimized implementations before it becomes evidence about cipher architecture.

3.3 Performance metric and scope

All practical measurements were obtained on an ASUSTeK ROG Strix G18 (board G815JPR) with an Intel Core i9-14900HX, BIOS G815JPR.318 dated 20 May 2026, microcode 0x133, Ubuntu 24.04.2 LTS, and kernel 6.17.0-22-generic. The exact compiler identification is gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0. The Makefile invokes cc, which resolves through the system alternatives to x86_64-linux-gnu-gcc-13. The complete common flags are -O3 -march=native -std=gnu11 -Wall -Wextra -Wpedantic -Iinclude -Isrc; on this compiler and processor, -march=native resolves to -march=alderlake -mtune=alderlake. AVX-512 is unavailable on this processor configuration, and AES-NI, GFNI, and VAES are not used by the measured kernels.

The benchmark process was pinned to logical CPU 2, a P-core, with `taskset -c 2`. The runs used a 64 KiB working set, three warm-up passes, and the median of 15 timed trials; the 128-bit benchmark uses the same protocol, with medians over repeated whole-program runs where noted. The pinned system used `intel_pstate` HWP, governor `powersave` with energy-performance preference `performance`, turbo enabled, minimum/maximum performance percentages 15/100, SMT enabled, AC platform profile `performance`, and AC power. Numeric package power limits such as PL1/PL2 were not captured. These settings were recorded after the benchmark rather than emitted alongside each CSV. The measurement-relevant repository revision is commit `f76c0d77b7c801273171676cb75aa323a3b4fbfa` of the public `PresentContinuous` repository [20]; the benchmarks were run shortly before that commit, whose measurement-relevant sources are byte-identical to the tested working tree.

The primary unit is nominal invariant-TSC ticks per processed byte, denoted `cyc/B`. These values are not literal core-clock cycles: the invariant TSC runs at a nominal rate while the core frequency was not pinned and turbo remained enabled. Their purpose is controlled relative comparison, especially within one process. The benchmark platform is more capable than the intended low-tier legitimate endpoint, yet much less parallel and specialized than the assumed adversarial setup. It

represents neither endpoint of the operational model.

Consequently, the measurements support claims about relative software cost and implementation structure, but not exact performance on Cortex-A7, Cortex-M, GPU, FPGA, or ASIC. Those platforms are discussed analytically or through published external results. A full implementation cycle on every platform is intentionally outside the present paper.

4 Representative construction approaches

The comparison set is deliberately small. It is selected to cover distinct construction approaches rather than to survey all lightweight cryptography.

4.1 Small S-box and bit permutation: PRESENT

PRESENT uses sixteen 4-bit S-boxes and a fixed bit permutation over a 64-bit state [21]. Its 4-bit S-box has a very small Boolean circuit, while the permutation can be free wiring in hardware and register renaming in bitslice. Although the direct scalar representation is awkward, optimized bitsliced software can be efficient even on embedded ARM processors [22]. The same features also benefit a massively parallel attacker.

4.2 Wide unstructured S-box and bit permutation: Donation

The evaluated DONATION instance follows the current unpublished specification [1]: eight copies of an 8-bit S-box, an 8×8 bit transpose, eight rounds, and final whitening. The S-box table in that manuscript is the table used by the software artifact; it supersedes the earlier workshop version for the purposes of this study. Its measured properties are differential uniformity 4, linearity 32, algebraic degree 7, and differential branch number 3. The transpose is an involutive bit permutation and contributes no XOR mixing.

The byte-oriented table representation is natural, but the best generic bitsliced circuit found for the supplied S-box contains 1117 gates in the AVX2 backend. This makes the fastest implementation table-driven and non-constant-time.

4.3 Wide S-box and XOR-and-rotate diffusion

The `lin444` mapping replaces a bit permutation by a unitriangular linear transformation over four words. It belongs to the cyclic-shift-and-XOR design family developed in earlier Qalqan-related work and its later generation methodology [23, 24]. The present 64-bit and 128-bit variants are experimental adaptations in the same family, not claims that the tested layer is identical to a deployed Qalqan round. For the 64-bit constructions the state is four 16-bit words; for the 128-bit construction it is four 32-bit words. Each word absorbs rotated versions of the others, and the inverse is obtained by undoing the four assignments in reverse order. Rotation constants are searched rather than assumed.

This approach can improve diffusion at low table-driven cost because a linear layer can be fused into precomputed substitution tables. In bitsliced software, however, its XOR network is explicit and may double the cost of a very cheap PRESENT round.

4.4 Structured 8-bit S-box: AES

AES supplies a useful control case: its S-box has the same differential uniformity, linearity, and algebraic degree as the supplied DONATION S-box, yet it has exploitable algebraic structure. A published Boyar–Peralta circuit is roughly an order of magnitude smaller than the generic BDD-derived circuit used for the supplied DONATION S-box [25]. The experiment therefore separates conventional cryptographic S-box metrics from implementation structure.

The 128-bit comparison includes reduced AES with MixColumns and an AES-S-box construction in which MixColumns is replaced by the selected `lin444` constants. These reduced ciphers are analytical round-function probes, not deployment recommendations.

4.5 Excluded architectures

PRINCE and ChaCha are relevant related work but are outside the primary experiment. PRINCE optimizes fully unrolled hardware latency and encryption/decryption reuse rather than the present ordinary-processor objective. ChaCha is an ARX stream cipher and does not admit the same active-S-box trail metric. Including either would require a second normalization framework and would obscure the present result.

5 Experimental results

5.1 Rounds required by the common yardstick

Table 4 reports the direct SAT decision results. “Witness” denotes a verified trail of weight below 64. “Proven” denotes an UNSAT result for weight at most 63.

Table 4: Minimum rounds reaching the 2^{-64} single-trail threshold and benchmark round count $X = r_{64} + 1$.

Construction	Witness below threshold	First proven round	r_{64}	X
PRESENT-80	14 rounds, weight 62	15	15	16
PRESENT- <code>lin444</code> -(2,9,7)	5 rounds, weight 63	6	6	7
DONATION	6 rounds, weight 60	7	7	8
DONATION- <code>lin444</code> -(2,9,7)	3 rounds, weight 63	4	4	5
DONATION- <code>lin444</code> -AES-S	3 rounds, weight 57	4	4	5
AES	3 rounds, weight 57	4	4	5
AES- <code>lin444</code> -(0,8,15)	2 rounds, weight 63	3	3	4

The PRESENT-`lin444` boundary is now exact. A verified five-round characteristic of weight 63 proves that five rounds remain below the threshold. A direct six-round decision instance for trails of weight at most 63 was proved UNSAT by CaDiCaL 3.0.1 after 20002 seconds in a randomized mixed-solver portfolio, and an independent fresh-seed rerun reproduced the UNSAT result after 20719 seconds. The SAT witness and both proof outcomes are retained in the artifact. Consequently, $r_{64} = 6$ and $X = 7$ for this construction.

5.2 Cost at equal differential-trail margin

Table 5 gives the best measured software implementation for each construction at X . Gaps below approximately 10% between the separate 64-bit and 128-bit benchmark programs are treated as

ties.

Table 5: Best software cost at $X = r_{64} + 1$. No AES-NI, GFNI, or VAES is used.

Construction	Block	X	Cost (cyc/B)	Winning kernel			Constant-time
PRESENT-1in444-(2,9,7)	64	7	0.502	AVX2	bitslice, 256 blocks		yes
PRESENT-80	64	16	0.553	AVX2	bitslice, 256 blocks		yes
DONATION-1in444-AES-S	64	5	0.868	AVX2	bitslice, 256 blocks		yes
AES-1in444-(0,8,15)	128	4	1.185	AVX2	bitslice, 256 blocks		yes
AES	128	5	1.269	AVX2	bitslice, 256 blocks		yes
DONATION-1in444	64	5	1.531	T-table, eight blocks			no
DONATION	64	8	2.415	T-table, eight blocks			no

The main findings are:

1. The round count alone does not predict cost. Per-round cost ranges from 0.035 cyc/B for PRESENT to about 0.306 cyc/B for the table-bound wide-S-box construction.
2. PRESENT-1in444 is the fastest measured equal-margin construction. Its round costs approximately twice as much as a PRESENT round, but it reaches the common yardstick in seven rounds instead of sixteen, resulting in 0.502 rather than 0.553 cyc/B—a 9% reduction on the measured platform.
3. The supplied DONATION S-box prevents an efficient bitslice path. Replacing only the linear layer improves the required round count from eight to five but leaves the kernel table-bound.
4. Replacing the supplied S-box by the AES S-box changes the winning implementation family and reduces the five-round cost from 1.531 to 0.868 cyc/B.
5. AES-1in444 reaches the threshold one round earlier than AES, but its more expensive round nearly cancels that gain. The 6.6% nominal difference is below the cross-program uncertainty and should be reported as an approximate tie.

5.3 Equal-margin cost versus differential-trail growth

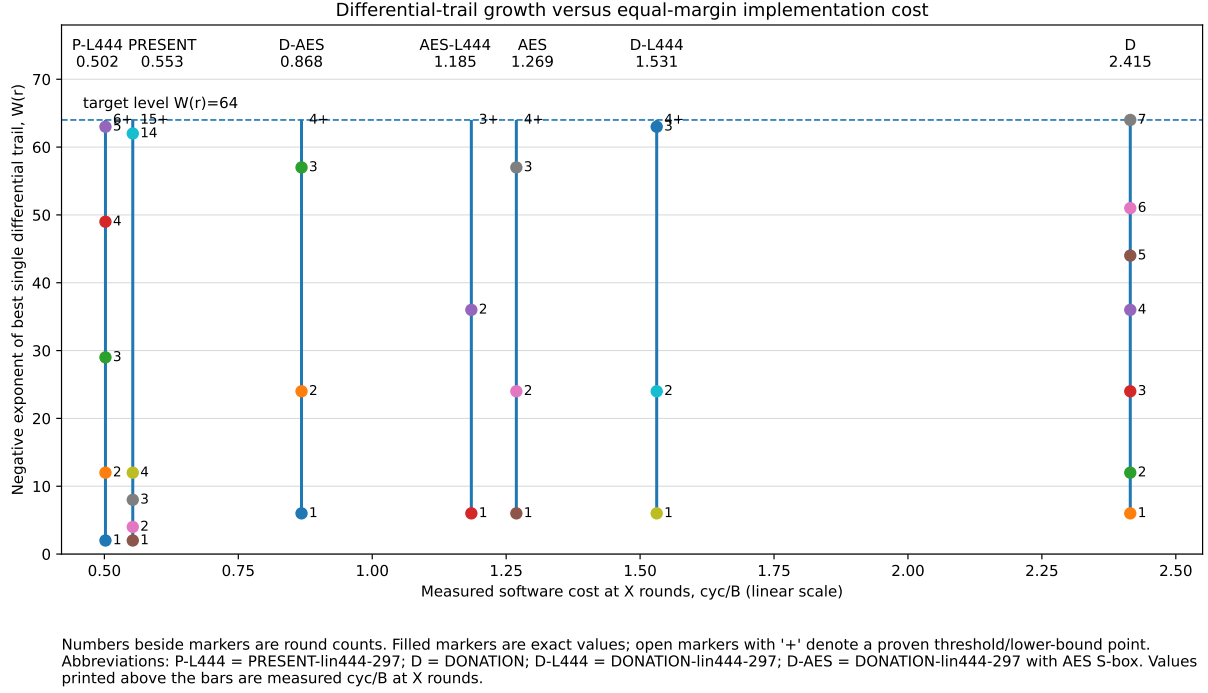
Figure 1 combines the two quantities directly supported by the experiment. The horizontal coordinate is the measured total software cost at the selected operating point $X = r_{64} + 1$, on a linear scale. Each vertical bar is one construction, and the markers show verified single-trail exponents or threshold evidence at selected round counts. Thus the figure answers a defender-side question: how much the normalized implementation costs and how quickly its differential-trail bound grows. It does not represent attacker hardware cost and therefore is not an AC/DC plot.

6 Donation case study

6.1 Specification under test

The evaluated cipher is the operationally supplied DONATION variant:

- 64-bit block;
- 576-bit raw master key interpreted as nine independent 64-bit round keys;



6.3 Replacing the transpose

Replacing only the bit transpose by `lin444` with constants (2, 9, 7) changes the exact weights for two, three, and four rounds from 12, 24, 36 to 24, 52, 68. Consequently, r_{64} falls from 7 to 4. At the original eight-round count, the fastest table implementation remains within measurement noise because the linear layer is fused into the tables. At the equal-margin count of five rounds, the cost falls from 2.415 to 1.531 cyc/B.

This ablation supports a narrow but useful construction principle: when a design already pays for an expensive wide S-box, a real XOR-linear layer can be nearly free in the fastest table path while buying substantially more differential diffusion.

6.4 Replacing the S-box

Replacing the supplied S-box by the AES S-box while keeping `lin444` and the same round-key model produces a second controlled ablation. The two S-boxes tie on differential uniformity, linearity, and algebraic degree; the supplied DONATION S-box has the better differential branch number. Nevertheless, the AES-S-box variant has a slightly better four-round trail bound in this particular layer and, more importantly, admits a compact published circuit.

The repository’s generic BDD synthesis produces about 1100 gates for either table. The known AES circuit reduces that to roughly 132 operations in the implemented accounting, changing the winning kernel from tables to AVX2 bitslice. At five rounds the cost falls to 0.868 cyc/B. This is still 1.57 times PRESENT-80 at the same yardstick, but it is 1.76 times faster than the corresponding supplied-S-box construction.

The result demonstrates that conventional S-box security metrics do not predict software circuit cost. Algebraic structure may improve legitimate constant-time performance, but it simultaneously provides the attacker with the same compact circuit and with potential specialized-instruction support.

6.5 Application assessment

The original DONATION design is not broken by the present single-trail analysis. It does, however, have four application-level disadvantages under the stated scenario:

1. a thin one-round single-trail margin in the original construction;
2. a supplied 8-bit S-box whose best known circuit is too large for efficient bitslicing;
3. a transpose that is weaker than an inexpensive XOR-linear alternative;
4. a 64-bit block, which creates a large generic distinguishing term over a 21.6 GB mission and requires strict counter-domain management.

Accordingly, the evidence does not support the claim that the original design is more efficient than PRESENT under a fair equal-margin software comparison. It also does not support declaring the cipher practically broken. The correct conclusion is conditional: DONATION passes the selected 2^{-64} single-trail screening rule at seven rounds, but its original architecture is dominated by its own controlled modifications and by reduced PRESENT on the measured platform.

7 Discussion

7.1 Which approach looks best?

Under the measured general-purpose software and single-trail model, the best current approach is a PRESENT-like small 4-bit S-box combined with a nontrivial XOR-and-rotate linear layer. The

layer doubles the measured round cost, but it reduces the equal-margin round count from sixteen to seven and yields the lowest total cost in the experiment. Thus the result favours spending more work on diffusion when the resulting reduction in rounds is larger than the per-round penalty. It is not an endorsement of 64-bit blocks for high-volume operation, nor evidence that this architecture is difficult for specialized attack hardware. The same small Boolean S-box and regular XOR network may also be inexpensive to replicate.

After the 12 Mbit/s, four-hour application filter is applied, a 128-bit block or a native stream construction is preferable. The current experiment does not include a 128-bit extension of the winning PRESENT-1in444 architecture, so it cannot determine whether that extension or a byte-oriented SPN is the better application-level design. Among the two tested 128-bit approaches, reduced AES and AES-1in444 are approximately tied: the latter buys one round of trail margin, while the former has the cheaper round and more mature implementation ecosystem.

The current evidence therefore supports the following ranked conclusions:

1. **Best measured equal-margin software cost:** PRESENT-1in444-(2,9,7), at $X = 7$ and 0.502 cyc/B, approximately 9% below PRESENT-80 at $X = 16$.
2. **Best current 128-bit approach:** reduced AES and AES-1in444 are approximately tied.
3. **Best lesson from Donation:** a wide S-box needs both a real diffusion layer and a compact implementation structure; reducing the round count alone is not enough.
4. **Unresolved operational winner:** no construction can yet be declared optimal for the actual endpoint-versus-attacker model because neither endpoint has been measured directly and the attack-cluster estimate remains incomplete.

7.2 The AC/DC objective remains unmeasured

The desired construction should maximize AC/DC: it should minimize the cost of protecting the mission stream while making mass replication and successful cryptanalysis expensive for the attacker. The present measurements reveal implementation sensitivity and determine the defender-side software ordering, but they do not provide C_A for the exact reduced variants. Consequently, they cannot produce a numerical AC/DC ranking. In particular:

- a tiny bitsliced S-box benefits both sides;
- a table-friendly wide S-box may slow constant-time software but remain cheap in FPGA ROM or ASIC logic;
- a structured S-box can dramatically improve both legitimate SIMD and attacker SIMD;
- an XOR-and-rotate layer is inexpensive for both sides;
- a costly one-time key expansion could create amortization asymmetry, but DONATION has no expansion and key expansion is outside the current experiment.

A construction change that appears free on the defender side is especially interesting in this framework. For example, replacing DONATION’s transpose by 1in444 is essentially free in its fastest table path while sharply improving the single-trail bound. Whether that change also raises attacker cost is unknown; an attacker may fuse the same layer or realize it cheaply in hardware. This illustrates why defender-side “free” diffusion is promising but not sufficient evidence of an AC/DC gain.

The quantitative result is therefore deliberately asymmetric in evidential strength. The GPU row in table 3 is a source-backed attack-capacity estimate. The CPU and V80 rows in table 2 establish only how much equipment fits within the capital bound; they do not determine Q_C . The historical COPACOBANA result confirms that PRESENT is exceptionally suitable for replicated FPGA search, but does not justify a direct forecast for modern cards. The ASIC class remains a sensitivity bound rather than a numerical result. This separation prevents fixed-key encryption throughput, device resource counts, and complete candidate-key throughput from being treated

as interchangeable quantities.

7.3 Implications for a later full design

A future design for the stated mission should probably use:

- a 128-bit block or a native stream-cipher construction;
- explicit comparison of a 128-bit small-S-box/XOR-linear architecture against structured byte-oriented SPNs;
- a compact constant-time S-box implementation appropriate to the chosen state geometry;
- a nontrivial XOR-linear layer selected by actual multi-round trail search rather than one-round diffusion heuristics;
- an explicit counter/nonce format that cannot wrap or overlap within the mission;
- sufficient key entropy independently of any computational-hardening mechanism;
- optionally, a separately analyzed session-key preparation stage if practical computational asymmetry is required.

This is a construction direction, not yet a cipher proposal.

8 Limitations and reproducibility

The present study has the following limitations.

1. The security normalization uses only the best single differential trail. It does not cover differential hulls or other attack classes.
2. The safety margin is exactly one additional round. It is a controlled experimental rule, much smaller than a production design margin.
3. The practical benchmark platform represents neither the intended legitimate endpoint nor the adversary’s aggregate hardware.
4. Hardware finite-field instructions are disabled in the main comparison, although an attacker may use them where applicable.
5. No GPU, FPGA, or ASIC kernel was developed in this work. GPU candidate rates are imported from peer-reviewed measurements; FPGA evidence is historical; ASIC capacity is only a sensitivity envelope.
6. The 64-bit and 128-bit data come from separate benchmark programs; differences below about 10% are not interpreted as decisive.
7. The fastest table implementations are not constant-time. Side channels are excluded from the threat model but remain relevant for deployment.
8. The normative DONATION source used here is an unpublished manuscript. Its revision and publication status must remain explicit, and the software artifact must retain the exact evaluated S-box table.
9. The avalanche experiment is an empirical local-bias check only. It does not establish keystream uniqueness, PRP/PRF security, differential-hull bounds, or resistance to structural attacks.
10. AC/DC is defined as the principal design objective, but it is not numerically evaluated because attacker costs for the exact reduced variants are unavailable.

The software artifact performs the following cross-checks before timing:

- AES known-answer testing against FIPS 197 [26];
- cross-checking table, interleaved, scalar, and bitsliced implementations against reference implementations;
- exhaustive verification of the implemented 8-bit S-box circuits over all 256 inputs;
- replay of SAT witnesses against the exact S-box DDT and linear-layer matrix;
- reproduction of known PRESENT active-S-box sequences.

The complete artifact is maintained in the public `EncryptedSilence/PresentContinuous` repository [20], which is intended to remain the single source of truth. The paper pins the measurement revision `f76c0d77b7c801273171676cb75aa323a3b4fbfa`; variant definitions are stored in `variants/`, raw speed and differential results in `results/`, the exact environment record is stored in `docs/measurement-environment.md`, and the principal reproduction commands are listed in section A. The avalanche CSV is a supplementary project artifact. GitHub releases will be archived automatically through Zenodo; the paper will cite the stable concept DOI together with the exact release tag and commit. **[PLACEHOLDER: Insert the Zenodo concept DOI and final GitHub release tag after the bootstrap archival release.]**

9 Conclusion

This paper proposed a practical methodology for comparing lightweight block-cipher construction approaches under a short-lived, time- and budget-bounded field scenario. The central objective is not defender throughput alone, but the attacker-cost/defender-cost ratio, AC/DC. The method separates four questions that are often conflated: the operational success criterion, the primitive-level differential-trail yardstick, the defender-side implementation cost, and the attacker-side cost required to reach the fixed success probability. Every construction is reduced to $X = r_{64} + 1$, receives the same applicable software optimization classes, and is measured in nominal TSC ticks per byte on one controlled platform.

The preliminary software result is now fully resolved for all seven rows. `PRESENT-1in444-(2,9,7)` has the lowest equal-margin cost: the stronger linear layer doubles the per-round cost relative to `PRESENT`, but reduces the required operating point from sixteen to seven rounds and therefore produces a measured 9% speed advantage. The original `DONATION` approach—a wide unstructured S-box, a transpose, and few rounds—does not outperform either `PRESENT` construction after fair normalization. Replacing the transpose by a genuine XOR-and-rotate layer substantially improves the differential-trail bound at negligible table-driven cost, while replacing the manuscript S-box by the structured AES S-box enables a compact constant-time bitsliced implementation. These controlled changes improve `DONATION`, but do not make it the measured software winner.

For the actual 12 Mbit/s, four-hour mission, the best construction approach is not simply the cheapest 64-bit round function. The 64-bit block creates a large generic birthday-scale distinguishing term and a tight counter namespace, although unique-counter CTR does not suffer the Sweet32 plaintext-recovery mechanism. Among the tested 128-bit designs, reduced AES and `AES-1in444` are approximately tied. The evidence therefore supports two separate conclusions: a small S-box plus stronger XOR diffusion is best under the measured equal-margin software model, while the application model favours a 128-bit state. A future comparison must combine those requirements by testing a 128-bit small-S-box/XOR-linear construction against the currently measured AES-like alternatives; the present data do not justify selecting either family as the final operational winner.

The adversary model now supports one quantitative conclusion: a \$1M point populated with 180 RTX 4090-class accelerators can perform about $2^{54.2}$ – $2^{56.0}$ complete full-cipher key checks in four hours for the published `PRESENT` and `AES` kernels. At $\varepsilon = 10^{-6}$ this implies approximately 75–76 effective key bits, while conservative 2^{60} and 2^{63} sensitivity envelopes imply 80 and 83 bits. A 96-bit effective key is therefore ample against exhaustive search in the stated classical model.

The main unresolved question is AC/DC. The current work measures the defender-side denominator and supplies external attacker anchors, but it cannot calculate the ratio for the exact reduced variants. The next stage must estimate or measure attacker cost for each applicable optimization class and determine which design changes increase C_A without increasing C_D . This is the logical continuation of the present result and a stricter objective than identifying the fastest reduced cipher. The paper can state which construction approach looks better under the measured software,

block-size, and exhaustive-search models, but not that any proposed reduced cipher definitively maximizes AC/DC or defeats every specialized implementation of the modeled adversary.

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A Key experimental commands

```
# Repository revision used in this paper:
# f76c0d77b7c801273171676cb75aa323a3b4fbfa
# Compiler: gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
```

```
# Makefile CFLAGS:
# -O3 -march=native -std=gnu11 -Wall -Wextra -Wpedantic -Iinclude -Isrc
make
make test

taskset -c 2 ./build/bench --csv results/speed.csv
taskset -c 2 ./build/wide_bench --rounds 4 --c0 0 8 15 \
    --csv results/wide-speed-r4.csv

python3 analysis/prove_bound.py --variant present-80 \
    --rounds 15 --weight 64 --csv results/rounds-at-64.csv

python3 analysis/prove_bound.py --variants-dir variants/wide \
    --variant aes-lin444-0-8-15 --rounds 3 --weight 64
```